

82 LL GPIO Generic Driver

82.1 GPIO Firmware driver registers structures

82.1.1 LL_GPIO_InitTypeDef

LL_GPIO_InitTypeDef is defined in the `stm32f4xx_ll_gpio.h`

Data Fields

- *uint32_t* Pin
- *uint32_t* Mode
- *uint32_t* Speed
- *uint32_t* OutputType
- *uint32_t* Pull
- *uint32_t* Alternate

Field Documentation

- *uint32_t* **LL_GPIO_InitTypeDef::Pin**
Specifies the GPIO pins to be configured. This parameter can be any value of [GPIO_LL_EC_PIN](#)
- *uint32_t* **LL_GPIO_InitTypeDef::Mode**
Specifies the operating mode for the selected pins. This parameter can be a value of [GPIO_LL_EC_MODE](#).GPIO HW configuration can be modified afterwards using unitary function `LL_GPIO_SetPinMode()`.
- *uint32_t* **LL_GPIO_InitTypeDef::Speed**
Specifies the speed for the selected pins. This parameter can be a value of [GPIO_LL_EC_SPEED](#).GPIO HW configuration can be modified afterwards using unitary function `LL_GPIO_SetPinSpeed()`.
- *uint32_t* **LL_GPIO_InitTypeDef::OutputType**
Specifies the operating output type for the selected pins. This parameter can be a value of [GPIO_LL_EC_OUTPUT](#).GPIO HW configuration can be modified afterwards using unitary function `LL_GPIO_SetPinOutputType()`.
- *uint32_t* **LL_GPIO_InitTypeDef::Pull**
Specifies the operating Pull-up/Pull down for the selected pins. This parameter can be a value of [GPIO_LL_EC_PULL](#).GPIO HW configuration can be modified afterwards using unitary function `LL_GPIO_SetPinPull()`.
- *uint32_t* **LL_GPIO_InitTypeDef::Alternate**
Specifies the Peripheral to be connected to the selected pins. This parameter can be a value of [GPIO_LL_EC_AF](#).GPIO HW configuration can be modified afterwards using unitary function `LL_GPIO_SetAFPin_0_7()` and `LL_GPIO_SetAFPin_8_15()`.

82.2 GPIO Firmware driver API description

The following section lists the various functions of the GPIO library.

82.2.1 Detailed description of functions

LL_GPIO_SetPinMode

Function name

```
__STATIC_INLINE void LL_GPIO_SetPinMode (GPIO_TypeDef * GPIOx, uint32_t Pin, uint32_t Mode)
```

Function description

Configure gpio mode for a dedicated pin on dedicated port.

Parameters

- **GPIOx:** GPIO Port
- **Pin:** This parameter can be one of the following values:
 - LL_GPIO_PIN_0
 - LL_GPIO_PIN_1
 - LL_GPIO_PIN_2
 - LL_GPIO_PIN_3
 - LL_GPIO_PIN_4
 - LL_GPIO_PIN_5
 - LL_GPIO_PIN_6
 - LL_GPIO_PIN_7
 - LL_GPIO_PIN_8
 - LL_GPIO_PIN_9
 - LL_GPIO_PIN_10
 - LL_GPIO_PIN_11
 - LL_GPIO_PIN_12
 - LL_GPIO_PIN_13
 - LL_GPIO_PIN_14
 - LL_GPIO_PIN_15
- **Mode:** This parameter can be one of the following values:
 - LL_GPIO_MODE_INPUT
 - LL_GPIO_MODE_OUTPUT
 - LL_GPIO_MODE_ALTERNATE
 - LL_GPIO_MODE_ANALOG

Return values

- **None:**

Notes

- I/O mode can be Input mode, General purpose output, Alternate function mode or Analog.
- Warning: only one pin can be passed as parameter.

Reference Manual to LL API cross reference:

- MODER MODEy LL_GPIO_SetPinMode

LL_GPIO_GetPinMode

Function name

__STATIC_INLINE uint32_t LL_GPIO_GetPinMode (GPIO_TypeDef * GPIOx, uint32_t Pin)

Function description

Return gpio mode for a dedicated pin on dedicated port.

Parameters

- **GPIOx:** GPIO Port
- **Pin:** This parameter can be one of the following values:
 - LL_GPIO_PIN_0
 - LL_GPIO_PIN_1
 - LL_GPIO_PIN_2
 - LL_GPIO_PIN_3
 - LL_GPIO_PIN_4
 - LL_GPIO_PIN_5
 - LL_GPIO_PIN_6
 - LL_GPIO_PIN_7
 - LL_GPIO_PIN_8
 - LL_GPIO_PIN_9
 - LL_GPIO_PIN_10
 - LL_GPIO_PIN_11
 - LL_GPIO_PIN_12
 - LL_GPIO_PIN_13
 - LL_GPIO_PIN_14
 - LL_GPIO_PIN_15

Return values

- **Returned:** value can be one of the following values:
 - LL_GPIO_MODE_INPUT
 - LL_GPIO_MODE_OUTPUT
 - LL_GPIO_MODE_ALTERNATE
 - LL_GPIO_MODE_ANALOG

Notes

- I/O mode can be Input mode, General purpose output, Alternate function mode or Analog.
- Warning: only one pin can be passed as parameter.

Reference Manual to LL API cross reference:

- MODER MODEy LL_GPIO_GetPinMode

LL_GPIO_SetPinOutputType

Function name

```
__STATIC_INLINE void LL_GPIO_SetPinOutputType (GPIO_TypeDef * GPIOx, uint32_t PinMask, uint32_t OutputType)
```

Function description

Configure gpio output type for several pins on dedicated port.

Parameters

- **GPIOx:** GPIO Port
- **PinMask:** This parameter can be a combination of the following values:
 - LL_GPIO_PIN_0
 - LL_GPIO_PIN_1
 - LL_GPIO_PIN_2
 - LL_GPIO_PIN_3
 - LL_GPIO_PIN_4
 - LL_GPIO_PIN_5
 - LL_GPIO_PIN_6
 - LL_GPIO_PIN_7
 - LL_GPIO_PIN_8
 - LL_GPIO_PIN_9
 - LL_GPIO_PIN_10
 - LL_GPIO_PIN_11
 - LL_GPIO_PIN_12
 - LL_GPIO_PIN_13
 - LL_GPIO_PIN_14
 - LL_GPIO_PIN_15
 - LL_GPIO_PIN_ALL
- **OutputType:** This parameter can be one of the following values:
 - LL_GPIO_OUTPUT_PUSH_PULL
 - LL_GPIO_OUTPUT_OPENDRAIN

Return values

- **None:**

Notes

- Output type as to be set when gpio pin is in output or alternate modes. Possible type are Push-pull or Open-drain.

Reference Manual to LL API cross reference:

- OTYPER OTy LL_GPIO_SetPinOutputType

LL_GPIO_GetPinOutputType

Function name

__STATIC_INLINE uint32_t LL_GPIO_GetPinOutputType (GPIO_TypeDef * GPIOx, uint32_t Pin)

Function description

Return gpio output type for several pins on dedicated port.

Parameters

- **GPIOx:** GPIO Port
- **Pin:** This parameter can be one of the following values:
 - LL_GPIO_PIN_0
 - LL_GPIO_PIN_1
 - LL_GPIO_PIN_2
 - LL_GPIO_PIN_3
 - LL_GPIO_PIN_4
 - LL_GPIO_PIN_5
 - LL_GPIO_PIN_6
 - LL_GPIO_PIN_7
 - LL_GPIO_PIN_8
 - LL_GPIO_PIN_9
 - LL_GPIO_PIN_10
 - LL_GPIO_PIN_11
 - LL_GPIO_PIN_12
 - LL_GPIO_PIN_13
 - LL_GPIO_PIN_14
 - LL_GPIO_PIN_15
 - LL_GPIO_PIN_ALL

Return values

- **Returned:** value can be one of the following values:
 - LL_GPIO_OUTPUT_PUSHPULL
 - LL_GPIO_OUTPUT_OPENDRAIN

Notes

- Output type as to be set when gpio pin is in output or alternate modes. Possible type are Push-pull or Open-drain.
- Warning: only one pin can be passed as parameter.

Reference Manual to LL API cross reference:

- OTYPER OTy LL_GPIO_GetPinOutputType

LL_GPIO_SetPinSpeed

Function name

__STATIC_INLINE void LL_GPIO_SetPinSpeed (GPIO_TypeDef * GPIOx, uint32_t Pin, uint32_t Speed)

Function description

Configure gpio speed for a dedicated pin on dedicated port.

Parameters

- **GPIOx:** GPIO Port
- **Pin:** This parameter can be one of the following values:
 - LL_GPIO_PIN_0
 - LL_GPIO_PIN_1
 - LL_GPIO_PIN_2
 - LL_GPIO_PIN_3
 - LL_GPIO_PIN_4
 - LL_GPIO_PIN_5
 - LL_GPIO_PIN_6
 - LL_GPIO_PIN_7
 - LL_GPIO_PIN_8
 - LL_GPIO_PIN_9
 - LL_GPIO_PIN_10
 - LL_GPIO_PIN_11
 - LL_GPIO_PIN_12
 - LL_GPIO_PIN_13
 - LL_GPIO_PIN_14
 - LL_GPIO_PIN_15
- **Speed:** This parameter can be one of the following values:
 - LL_GPIO_SPEED_FREQ_LOW
 - LL_GPIO_SPEED_FREQ_MEDIUM
 - LL_GPIO_SPEED_FREQ_HIGH
 - LL_GPIO_SPEED_FREQ_VERY_HIGH

Return values

- **None:**

Notes

- I/O speed can be Low, Medium, Fast or High speed.
- Warning: only one pin can be passed as parameter.
- Refer to datasheet for frequency specifications and the power supply and load conditions for each speed.

Reference Manual to LL API cross reference:

- OSPEEDR OSPEEDy LL_GPIO_SetPinSpeed

LL_GPIO_GetPinSpeed

Function name

```
__STATIC_INLINE uint32_t LL_GPIO_GetPinSpeed(GPIO_TypeDef * GPIOx, uint32_t Pin)
```

Function description

Return gpio speed for a dedicated pin on dedicated port.

Parameters

- **GPIOx:** GPIO Port
- **Pin:** This parameter can be one of the following values:
 - LL_GPIO_PIN_0
 - LL_GPIO_PIN_1
 - LL_GPIO_PIN_2
 - LL_GPIO_PIN_3
 - LL_GPIO_PIN_4
 - LL_GPIO_PIN_5
 - LL_GPIO_PIN_6
 - LL_GPIO_PIN_7
 - LL_GPIO_PIN_8
 - LL_GPIO_PIN_9
 - LL_GPIO_PIN_10
 - LL_GPIO_PIN_11
 - LL_GPIO_PIN_12
 - LL_GPIO_PIN_13
 - LL_GPIO_PIN_14
 - LL_GPIO_PIN_15

Return values

- **Returned:** value can be one of the following values:
 - LL_GPIO_SPEED_FREQ_LOW
 - LL_GPIO_SPEED_FREQ_MEDIUM
 - LL_GPIO_SPEED_FREQ_HIGH
 - LL_GPIO_SPEED_FREQ_VERY_HIGH

Notes

- I/O speed can be Low, Medium, Fast or High speed.
- Warning: only one pin can be passed as parameter.
- Refer to datasheet for frequency specifications and the power supply and load conditions for each speed.

Reference Manual to LL API cross reference:

- OSPEEDR OSPEEDy LL_GPIO_GetPinSpeed

LL_GPIO_SetPinPull

Function name

__STATIC_INLINE void LL_GPIO_SetPinPull (GPIO_TypeDef * GPIOx, uint32_t Pin, uint32_t Pull)

Function description

Configure gpio pull-up or pull-down for a dedicated pin on a dedicated port.

Parameters

- **GPIOx:** GPIO Port
- **Pin:** This parameter can be one of the following values:
 - LL_GPIO_PIN_0
 - LL_GPIO_PIN_1
 - LL_GPIO_PIN_2
 - LL_GPIO_PIN_3
 - LL_GPIO_PIN_4
 - LL_GPIO_PIN_5
 - LL_GPIO_PIN_6
 - LL_GPIO_PIN_7
 - LL_GPIO_PIN_8
 - LL_GPIO_PIN_9
 - LL_GPIO_PIN_10
 - LL_GPIO_PIN_11
 - LL_GPIO_PIN_12
 - LL_GPIO_PIN_13
 - LL_GPIO_PIN_14
 - LL_GPIO_PIN_15
- **Pull:** This parameter can be one of the following values:
 - LL_GPIO_PULL_NO
 - LL_GPIO_PULL_UP
 - LL_GPIO_PULL_DOWN

Return values

- **None:**

Notes

- Warning: only one pin can be passed as parameter.

Reference Manual to LL API cross reference:

- PUPDR PUPDy LL_GPIO_SetPinPull

LL_GPIO_GetPinPull

Function name

__STATIC_INLINE uint32_t LL_GPIO_GetPinPull (GPIO_TypeDef * GPIOx, uint32_t Pin)

Function description

Return gpio pull-up or pull-down for a dedicated pin on a dedicated port.

Parameters

- **GPIOx:** GPIO Port
- **Pin:** This parameter can be one of the following values:
 - LL_GPIO_PIN_0
 - LL_GPIO_PIN_1
 - LL_GPIO_PIN_2
 - LL_GPIO_PIN_3
 - LL_GPIO_PIN_4
 - LL_GPIO_PIN_5
 - LL_GPIO_PIN_6
 - LL_GPIO_PIN_7
 - LL_GPIO_PIN_8
 - LL_GPIO_PIN_9
 - LL_GPIO_PIN_10
 - LL_GPIO_PIN_11
 - LL_GPIO_PIN_12
 - LL_GPIO_PIN_13
 - LL_GPIO_PIN_14
 - LL_GPIO_PIN_15

Return values

- **Returned:** value can be one of the following values:
 - LL_GPIO_PULL_NO
 - LL_GPIO_PULL_UP
 - LL_GPIO_PULL_DOWN

Notes

- Warning: only one pin can be passed as parameter.

Reference Manual to LL API cross reference:

- PUPDR PUPDy LL_GPIO_GetPinPull

LL_GPIO_SetAFPin_0_7

Function name

__STATIC_INLINE void LL_GPIO_SetAFPin_0_7 (GPIO_TypeDef * GPIOx, uint32_t Pin, uint32_t Alternate)

Function description

Configure gpio alternate function of a dedicated pin from 0 to 7 for a dedicated port.

Parameters

- **GPIOx:** GPIO Port
- **Pin:** This parameter can be one of the following values:
 - LL_GPIO_PIN_0
 - LL_GPIO_PIN_1
 - LL_GPIO_PIN_2
 - LL_GPIO_PIN_3
 - LL_GPIO_PIN_4
 - LL_GPIO_PIN_5
 - LL_GPIO_PIN_6
 - LL_GPIO_PIN_7
- **Alternate:** This parameter can be one of the following values:
 - LL_GPIO_AF_0
 - LL_GPIO_AF_1
 - LL_GPIO_AF_2
 - LL_GPIO_AF_3
 - LL_GPIO_AF_4
 - LL_GPIO_AF_5
 - LL_GPIO_AF_6
 - LL_GPIO_AF_7
 - LL_GPIO_AF_8
 - LL_GPIO_AF_9
 - LL_GPIO_AF_10
 - LL_GPIO_AF_11
 - LL_GPIO_AF_12
 - LL_GPIO_AF_13
 - LL_GPIO_AF_14
 - LL_GPIO_AF_15

Return values

- **None:**

Notes

- Possible values are from AF0 to AF15 depending on target.
- Warning: only one pin can be passed as parameter.

Reference Manual to LL API cross reference:

- AFRL AFSELY LL_GPIO_SetAFPin_0_7

LL_GPIO_GetAFPin_0_7

Function name

`__STATIC_INLINE uint32_t LL_GPIO_GetAFPin_0_7 (GPIO_TypeDef * GPIOx, uint32_t Pin)`

Function description

Return gpio alternate function of a dedicated pin from 0 to 7 for a dedicated port.

Parameters

- **GPIOx:** GPIO Port
- **Pin:** This parameter can be one of the following values:
 - LL_GPIO_PIN_0
 - LL_GPIO_PIN_1
 - LL_GPIO_PIN_2
 - LL_GPIO_PIN_3
 - LL_GPIO_PIN_4
 - LL_GPIO_PIN_5
 - LL_GPIO_PIN_6
 - LL_GPIO_PIN_7

Return values

- **Returned:** value can be one of the following values:
 - LL_GPIO_AF_0
 - LL_GPIO_AF_1
 - LL_GPIO_AF_2
 - LL_GPIO_AF_3
 - LL_GPIO_AF_4
 - LL_GPIO_AF_5
 - LL_GPIO_AF_6
 - LL_GPIO_AF_7
 - LL_GPIO_AF_8
 - LL_GPIO_AF_9
 - LL_GPIO_AF_10
 - LL_GPIO_AF_11
 - LL_GPIO_AF_12
 - LL_GPIO_AF_13
 - LL_GPIO_AF_14
 - LL_GPIO_AF_15

Reference Manual to LL API cross reference:

- AFRL AFSELY LL_GPIO_GetAFPin_0_7

LL_GPIO_SetAFPin_8_15

Function name

```
__STATIC_INLINE void LL_GPIO_SetAFPin_8_15 (GPIO_TypeDef * GPIOx, uint32_t Pin, uint32_t Alternate)
```

Function description

Configure gpio alternate function of a dedicated pin from 8 to 15 for a dedicated port.

Parameters

- **GPIOx:** GPIO Port
- **Pin:** This parameter can be one of the following values:
 - LL_GPIO_PIN_8
 - LL_GPIO_PIN_9
 - LL_GPIO_PIN_10
 - LL_GPIO_PIN_11
 - LL_GPIO_PIN_12
 - LL_GPIO_PIN_13
 - LL_GPIO_PIN_14
 - LL_GPIO_PIN_15
- **Alternate:** This parameter can be one of the following values:
 - LL_GPIO_AF_0
 - LL_GPIO_AF_1
 - LL_GPIO_AF_2
 - LL_GPIO_AF_3
 - LL_GPIO_AF_4
 - LL_GPIO_AF_5
 - LL_GPIO_AF_6
 - LL_GPIO_AF_7
 - LL_GPIO_AF_8
 - LL_GPIO_AF_9
 - LL_GPIO_AF_10
 - LL_GPIO_AF_11
 - LL_GPIO_AF_12
 - LL_GPIO_AF_13
 - LL_GPIO_AF_14
 - LL_GPIO_AF_15

Return values

- **None:**

Notes

- Possible values are from AF0 to AF15 depending on target.
- Warning: only one pin can be passed as parameter.

Reference Manual to LL API cross reference:

- AFRH AFSELY LL_GPIO_SetAFPin_8_15

LL_GPIO_GetAFPin_8_15

Function name

__STATIC_INLINE uint32_t LL_GPIO_GetAFPin_8_15 (GPIO_TypeDef * GPIOx, uint32_t Pin)

Function description

Return gpio alternate function of a dedicated pin from 8 to 15 for a dedicated port.

Parameters

- **GPIOx:** GPIO Port
- **Pin:** This parameter can be one of the following values:
 - LL_GPIO_PIN_8
 - LL_GPIO_PIN_9
 - LL_GPIO_PIN_10
 - LL_GPIO_PIN_11
 - LL_GPIO_PIN_12
 - LL_GPIO_PIN_13
 - LL_GPIO_PIN_14
 - LL_GPIO_PIN_15

Return values

- **Returned:** value can be one of the following values:
 - LL_GPIO_AF_0
 - LL_GPIO_AF_1
 - LL_GPIO_AF_2
 - LL_GPIO_AF_3
 - LL_GPIO_AF_4
 - LL_GPIO_AF_5
 - LL_GPIO_AF_6
 - LL_GPIO_AF_7
 - LL_GPIO_AF_8
 - LL_GPIO_AF_9
 - LL_GPIO_AF_10
 - LL_GPIO_AF_11
 - LL_GPIO_AF_12
 - LL_GPIO_AF_13
 - LL_GPIO_AF_14
 - LL_GPIO_AF_15

Notes

- Possible values are from AF0 to AF15 depending on target.

Reference Manual to LL API cross reference:

- AFRH AFSELY LL_GPIO_GetAFPin_8_15

LL_GPIO_LockPin

Function name

```
__STATIC_INLINE void LL_GPIO_LockPin (GPIO_TypeDef * GPIOx, uint32_t PinMask)
```

Function description

Lock configuration of several pins for a dedicated port.

Parameters

- **GPIOx:** GPIO Port
- **PinMask:** This parameter can be a combination of the following values:
 - LL_GPIO_PIN_0
 - LL_GPIO_PIN_1
 - LL_GPIO_PIN_2
 - LL_GPIO_PIN_3
 - LL_GPIO_PIN_4
 - LL_GPIO_PIN_5
 - LL_GPIO_PIN_6
 - LL_GPIO_PIN_7
 - LL_GPIO_PIN_8
 - LL_GPIO_PIN_9
 - LL_GPIO_PIN_10
 - LL_GPIO_PIN_11
 - LL_GPIO_PIN_12
 - LL_GPIO_PIN_13
 - LL_GPIO_PIN_14
 - LL_GPIO_PIN_15
 - LL_GPIO_PIN_ALL

Return values

- **None:**

Notes

- When the lock sequence has been applied on a port bit, the value of this port bit can no longer be modified until the next reset.
- Each lock bit freezes a specific configuration register (control and alternate function registers).

Reference Manual to LL API cross reference:

- LCKR LCKK LL_GPIO_LockPin

LL_GPIO_IsPinLocked

Function name

```
__STATIC_INLINE uint32_t LL_GPIO_IsPinLocked (GPIO_TypeDef * GPIOx, uint32_t PinMask)
```

Function description

Return 1 if all pins passed as parameter, of a dedicated port, are locked.

Parameters

- **GPIOx:** GPIO Port
- **PinMask:** This parameter can be a combination of the following values:
 - LL_GPIO_PIN_0
 - LL_GPIO_PIN_1
 - LL_GPIO_PIN_2
 - LL_GPIO_PIN_3
 - LL_GPIO_PIN_4
 - LL_GPIO_PIN_5
 - LL_GPIO_PIN_6
 - LL_GPIO_PIN_7
 - LL_GPIO_PIN_8
 - LL_GPIO_PIN_9
 - LL_GPIO_PIN_10
 - LL_GPIO_PIN_11
 - LL_GPIO_PIN_12
 - LL_GPIO_PIN_13
 - LL_GPIO_PIN_14
 - LL_GPIO_PIN_15
 - LL_GPIO_PIN_ALL

Return values

- **State:** of bit (1 or 0).

Reference Manual to LL API cross reference:

- LCKR LCKy LL_GPIO_IsPinLocked

LL_GPIO_IsAnyPinLocked

Function name

```
__STATIC_INLINE uint32_t LL_GPIO_IsAnyPinLocked (GPIO_TypeDef * GPIOx)
```

Function description

Return 1 if one of the pin of a dedicated port is locked.

Parameters

- **GPIOx:** GPIO Port

Return values

- **State:** of bit (1 or 0).

Reference Manual to LL API cross reference:

- LCKR LCKK LL_GPIO_IsAnyPinLocked

LL_GPIO_ReadInputPort

Function name

```
__STATIC_INLINE uint32_t LL_GPIO_ReadInputPort (GPIO_TypeDef * GPIOx)
```

Function description

Return full input data register value for a dedicated port.

Parameters

- **GPIOx:** GPIO Port

Return values

- **Input:** data register value of port

Reference Manual to LL API cross reference:

- IDR IDy LL_GPIO_ReadInputPort

LL_GPIO_IsInputPinSet

Function name

```
__STATIC_INLINE uint32_t LL_GPIO_IsInputPinSet (GPIO_TypeDef * GPIOx, uint32_t PinMask)
```

Function description

Return if input data level for several pins of dedicated port is high or low.

Parameters

- **GPIOx:** GPIO Port
- **PinMask:** This parameter can be a combination of the following values:
 - LL_GPIO_PIN_0
 - LL_GPIO_PIN_1
 - LL_GPIO_PIN_2
 - LL_GPIO_PIN_3
 - LL_GPIO_PIN_4
 - LL_GPIO_PIN_5
 - LL_GPIO_PIN_6
 - LL_GPIO_PIN_7
 - LL_GPIO_PIN_8
 - LL_GPIO_PIN_9
 - LL_GPIO_PIN_10
 - LL_GPIO_PIN_11
 - LL_GPIO_PIN_12
 - LL_GPIO_PIN_13
 - LL_GPIO_PIN_14
 - LL_GPIO_PIN_15
 - LL_GPIO_PIN_ALL

Return values

- **State:** of bit (1 or 0).

Reference Manual to LL API cross reference:

- IDR IDy LL_GPIO_IsInputPinSet

LL_GPIO_WriteOutputPort

Function name

```
__STATIC_INLINE void LL_GPIO_WriteOutputPort (GPIO_TypeDef * GPIOx, uint32_t PortValue)
```

Function description

Write output data register for the port.

Parameters

- **GPIOx:** GPIO Port
- **PortValue:** Level value for each pin of the port

Return values

- **None:**

Reference Manual to LL API cross reference:

- ODR ODy LL_GPIO_WriteOutputPort

LL_GPIO_ReadOutputPort

Function name

```
__STATIC_INLINE uint32_t LL_GPIO_ReadOutputPort (GPIO_TypeDef * GPIOx)
```

Function description

Return full output data register value for a dedicated port.

Parameters

- **GPIOx:** GPIO Port

Return values

- **Output:** data register value of port

Reference Manual to LL API cross reference:

- ODR ODy LL_GPIO_ReadOutputPort

LL_GPIO_IsOutputPinSet

Function name

```
__STATIC_INLINE uint32_t LL_GPIO_IsOutputPinSet (GPIO_TypeDef * GPIOx, uint32_t PinMask)
```

Function description

Return if input data level for several pins of dedicated port is high or low.

Parameters

- **GPIOx:** GPIO Port
- **PinMask:** This parameter can be a combination of the following values:
 - LL_GPIO_PIN_0
 - LL_GPIO_PIN_1
 - LL_GPIO_PIN_2
 - LL_GPIO_PIN_3
 - LL_GPIO_PIN_4
 - LL_GPIO_PIN_5
 - LL_GPIO_PIN_6
 - LL_GPIO_PIN_7
 - LL_GPIO_PIN_8
 - LL_GPIO_PIN_9
 - LL_GPIO_PIN_10
 - LL_GPIO_PIN_11
 - LL_GPIO_PIN_12
 - LL_GPIO_PIN_13
 - LL_GPIO_PIN_14
 - LL_GPIO_PIN_15
 - LL_GPIO_PIN_ALL

Return values

- **State:** of bit (1 or 0).

Reference Manual to LL API cross reference:

- ODR ODy LL_GPIO_IsOutputPinSet

LL_GPIO_SetOutputPin

Function name

```
__STATIC_INLINE void LL_GPIO_SetOutputPin (GPIO_TypeDef * GPIOx, uint32_t PinMask)
```

Function description

Set several pins to high level on dedicated gpio port.

Parameters

- **GPIOx:** GPIO Port
- **PinMask:** This parameter can be a combination of the following values:
 - LL_GPIO_PIN_0
 - LL_GPIO_PIN_1
 - LL_GPIO_PIN_2
 - LL_GPIO_PIN_3
 - LL_GPIO_PIN_4
 - LL_GPIO_PIN_5
 - LL_GPIO_PIN_6
 - LL_GPIO_PIN_7
 - LL_GPIO_PIN_8
 - LL_GPIO_PIN_9
 - LL_GPIO_PIN_10
 - LL_GPIO_PIN_11
 - LL_GPIO_PIN_12
 - LL_GPIO_PIN_13
 - LL_GPIO_PIN_14
 - LL_GPIO_PIN_15
 - LL_GPIO_PIN_ALL

Return values

- **None:**

Reference Manual to LL API cross reference:

- BSRR BSy LL_GPIO_SetOutputPin

LL_GPIO_ResetOutputPin

Function name

```
__STATIC_INLINE void LL_GPIO_ResetOutputPin (GPIO_TypeDef * GPIOx, uint32_t PinMask)
```

Function description

Set several pins to low level on dedicated gpio port.

Parameters

- **GPIOx:** GPIO Port
- **PinMask:** This parameter can be a combination of the following values:
 - LL_GPIO_PIN_0
 - LL_GPIO_PIN_1
 - LL_GPIO_PIN_2
 - LL_GPIO_PIN_3
 - LL_GPIO_PIN_4
 - LL_GPIO_PIN_5
 - LL_GPIO_PIN_6
 - LL_GPIO_PIN_7
 - LL_GPIO_PIN_8
 - LL_GPIO_PIN_9
 - LL_GPIO_PIN_10
 - LL_GPIO_PIN_11
 - LL_GPIO_PIN_12
 - LL_GPIO_PIN_13
 - LL_GPIO_PIN_14
 - LL_GPIO_PIN_15
 - LL_GPIO_PIN_ALL

Return values

- **None:**

Reference Manual to LL API cross reference:

- BSRR BRy LL_GPIO_ResetOutputPin

LL_GPIO_TogglePin

Function name

```
__STATIC_INLINE void LL_GPIO_TogglePin (GPIO_TypeDef * GPIOx, uint32_t PinMask)
```

Function description

Toggle data value for several pin of dedicated port.

Parameters

- **GPIOx:** GPIO Port
- **PinMask:** This parameter can be a combination of the following values:
 - LL_GPIO_PIN_0
 - LL_GPIO_PIN_1
 - LL_GPIO_PIN_2
 - LL_GPIO_PIN_3
 - LL_GPIO_PIN_4
 - LL_GPIO_PIN_5
 - LL_GPIO_PIN_6
 - LL_GPIO_PIN_7
 - LL_GPIO_PIN_8
 - LL_GPIO_PIN_9
 - LL_GPIO_PIN_10
 - LL_GPIO_PIN_11
 - LL_GPIO_PIN_12
 - LL_GPIO_PIN_13
 - LL_GPIO_PIN_14
 - LL_GPIO_PIN_15
 - LL_GPIO_PIN_ALL

Return values

- **None:**

Reference Manual to LL API cross reference:

- ODR ODy LL_GPIO_TogglePin

LL_GPIO_DeInit

Function name

ErrorStatus LL_GPIO_DeInit (GPIO_TypeDef * GPIOx)

Function description

De-initialize GPIO registers (Registers restored to their default values).

Parameters

- **GPIOx:** GPIO Port

Return values

- **An:** ErrorStatus enumeration value:
 - SUCCESS: GPIO registers are de-initialized
 - ERROR: Wrong GPIO Port

LL_GPIO_Init

Function name

ErrorStatus LL_GPIO_Init (GPIO_TypeDef * GPIOx, LL_GPIO_InitTypeDef * GPIO_InitStruct)

Function description

Initialize GPIO registers according to the specified parameters in GPIO_InitStruct.

Parameters

- **GPIOx:** GPIO Port
- **GPIO_InitStruct:** pointer to a LL_GPIO_InitTypeDef structure that contains the configuration information for the specified GPIO peripheral.

Return values

- **An:** ErrorStatus enumeration value:
 - SUCCESS: GPIO registers are initialized according to GPIO_InitStruct content
 - ERROR: Not applicable

LL_GPIO_StructInit

Function name

```
void LL_GPIO_StructInit (LL_GPIO_InitTypeDef * GPIO_InitStruct)
```

Function description

Set each LL_GPIO_InitTypeDef field to default value.

Parameters

- **GPIO_InitStruct:** pointer to a LL_GPIO_InitTypeDef structure whose fields will be set to default values.

Return values

- **None:**

82.3 GPIO Firmware driver defines

The following section lists the various define and macros of the module.

82.3.1 GPIO

GPIO

Alternate Function

LL_GPIO_AF_0

Select alternate function 0

LL_GPIO_AF_1

Select alternate function 1

LL_GPIO_AF_2

Select alternate function 2

LL_GPIO_AF_3

Select alternate function 3

LL_GPIO_AF_4

Select alternate function 4

LL_GPIO_AF_5

Select alternate function 5

LL_GPIO_AF_6

Select alternate function 6

LL_GPIO_AF_7

Select alternate function 7

LL_GPIO_AF_8

Select alternate function 8

LL_GPIO_AF_9

Select alternate function 9

LL_GPIO_AF_10

Select alternate function 10

LL_GPIO_AF_11

Select alternate function 11

LL_GPIO_AF_12

Select alternate function 12

LL_GPIO_AF_13

Select alternate function 13

LL_GPIO_AF_14

Select alternate function 14

LL_GPIO_AF_15

Select alternate function 15

Mode**LL_GPIO_MODE_INPUT**

Select input mode

LL_GPIO_MODE_OUTPUT

Select output mode

LL_GPIO_MODE_ALTERNATE

Select alternate function mode

LL_GPIO_MODE_ANALOG

Select analog mode

Output Type**LL_GPIO_OUTPUT_PUSHPULL**

Select push-pull as output type

LL_GPIO_OUTPUT_OPENDRAIN

Select open-drain as output type

PIN**LL_GPIO_PIN_0**

Select pin 0

LL_GPIO_PIN_1

Select pin 1

LL_GPIO_PIN_2

Select pin 2

LL_GPIO_PIN_3

Select pin 3

LL_GPIO_PIN_4

Select pin 4

LL_GPIO_PIN_5

Select pin 5

LL_GPIO_PIN_6

Select pin 6

LL_GPIO_PIN_7

Select pin 7

LL_GPIO_PIN_8

Select pin 8

LL_GPIO_PIN_9

Select pin 9

LL_GPIO_PIN_10

Select pin 10

LL_GPIO_PIN_11

Select pin 11

LL_GPIO_PIN_12

Select pin 12

LL_GPIO_PIN_13

Select pin 13

LL_GPIO_PIN_14

Select pin 14

LL_GPIO_PIN_15

Select pin 15

LL_GPIO_PIN_ALL

Select all pins

Pull Up Pull Down**LL_GPIO_PULL_NO**

Select I/O no pull

LL_GPIO_PULL_UP

Select I/O pull up

LL_GPIO_PULL_DOWN

Select I/O pull down

Output Speed**LL_GPIO_SPEED_FREQ_LOW**

Select I/O low output speed

LL_GPIO_SPEED_FREQ_MEDIUM

Select I/O medium output speed

LL_GPIO_SPEED_FREQ_HIGH

Select I/O fast output speed

LL_GPIO_SPEED_FREQ_VERY_HIGH

Select I/O high output speed

Common Write and read registers Macros**LL_GPIO_WriteReg****Description:**

- Write a value in GPIO register.

Parameters:

- `__INSTANCE__`: GPIO Instance
- `__REG__`: Register to be written
- `__VALUE__`: Value to be written in the register

Return value:

- None

LL_GPIO_ReadReg**Description:**

- Read a value in GPIO register.

Parameters:

- `__INSTANCE__`: GPIO Instance
- `__REG__`: Register to be read

Return value:

- Register: value